

A six-term progression in a cap-six additive basis modulo 29

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Abstract

We exhibit a subset of $\mathbb{Z}/29\mathbb{Z}$ containing six distinct terms in arithmetic progression and having between one and six ordered representations of every residue as a sum of two elements. A complete finite certificate verifies the representation counts. Together with the known endpoint-pair bound, this shows that the maximum progression length under the fixed representation cap six is exactly six. Using the previously reported value $R_{29} = 6$, the equality is $L_{29} = 6$ in the notation for extremal Ruzsa bases.

1 The fixed-cap result

Write $G = \mathbb{Z}/29\mathbb{Z}$ and, for $A \subseteq G$, define

$$\sigma_A(r) = \#\{(a, b) \in A \times A : a + b = r \text{ in } G\}.$$

The pairs are ordered, and diagonal pairs are included. Let

$$\mathcal{F}_6 = \{A \subseteq G : 1 \leq \sigma_A(r) \leq 6 \text{ for every } r \in G\}.$$

For a nonempty A , let $\ell(A)$ be the greatest k for which $x, x + d, \dots, x + (k - 1)d$ are distinct elements of A , for some $x, d \in G$. Modular wraparound is allowed; for $k \geq 2$, distinctness requires $d \neq 0$. There is no cardinality restriction on members of $\mathcal{F}_6$.

Theorem 1. *The fixed-cap maximum is*

$$\max_{A \in \mathcal{F}_6} \ell(A) = 6.$$

It is attained by

$$A_* = \{0, 1, 2, 3, 4, 5, 8, 13, 19, 23\}. \quad (1)$$

In particular, the assertion that this maximum equals five is false.

Proof. We reproduce the endpoint-pair argument of Ding, Li, Li, Niu and Zhao [1, Section 5]. Suppose $p_i = x + id$, $0 \leq i < k$, are distinct members of $A \in \mathcal{F}_6$. The ordered pairs

$$(p_i, p_{k-1-i}), \quad 0 \leq i < k,$$

are distinct because their first coordinates are distinct. Each has sum $2x + (k - 1)d$. Hence

$$k \leq \sigma_A(2x + (k - 1)d) \leq 6.$$

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Reversed pairs count separately, and a central diagonal pair, if present, counts once. This argument applies without change to progressions that wrap around the modulus.

For the reverse inequality, Table 1 lists all unordered pairs with repetition from A_* . A pair with unequal entries contributes two ordered representations, and a diagonal pair contributes one. The resulting counts, in residue order $0, \dots, 28$, are

$$\begin{aligned} &(1, 2, 5, 6, 5, 6, 5, 6, 5, 5, 3, 2, 2, 6, 2, \\ &2, 3, 3, 2, 2, 2, 4, 2, 4, 4, 2, 3, 4, 2). \end{aligned} \tag{2}$$

Every entry is between one and six, so $A_* \in \mathcal{F}_6$. Its elements $0, 1, 2, 3, 4, 5$ form a six-term progression. The family is therefore nonempty, and these bounds prove the stated maximum. $\square$

2 The finite certificate

Table 1: All pairs (a, b) from A_* with $a \leq b$, grouped by $r \equiv a + b \pmod{29}$. The column c is the ordered count $\sigma_{A_*}(r)$.

r	Pairs	c	r	Pairs	c
0	(0, 0)	1	15	(2, 13)	2
1	(0, 1)	2	16	(3, 13), (8, 8)	3
2	(0, 2), (1, 1), (8, 23)	5	17	(4, 13), (23, 23)	3
3	(0, 3), (1, 2), (13, 19)	6	18	(5, 13)	2
4	(0, 4), (1, 3), (2, 2)	5	19	(0, 19)	2
5	(0, 5), (1, 4), (2, 3)	6	20	(1, 19)	2
6	(1, 5), (2, 4), (3, 3)	5	21	(2, 19), (8, 13)	4
7	(2, 5), (3, 4), (13, 23)	6	22	(3, 19)	2
8	(0, 8), (3, 5), (4, 4)	5	23	(0, 23), (4, 19)	4
9	(1, 8), (4, 5), (19, 19)	5	24	(1, 23), (5, 19)	4
10	(2, 8), (5, 5)	3	25	(2, 23)	2
11	(3, 8)	2	26	(3, 23), (13, 13)	3
12	(4, 8)	2	27	(4, 23), (8, 19)	4
13	(0, 13), (5, 8), (19, 23)	6	28	(5, 23)	2
14	(1, 13)	2			

The table contains all $10 \cdot 11/2 = 55$ unordered pairs with repetition; their weights sum to $100 = |A_*|^2$. In particular, the six ordered representations of residue five are

$$(0, 5), (1, 4), (2, 3), (3, 2), (4, 1), (5, 0).$$

Thus the progression itself saturates the representation cap at this residue. The following complete Python 3 calculation reproduces (2) with exact integers:

```
A = (0, 1, 2, 3, 4, 5, 8, 13, 19, 23)
c = [sum((a + b) % 29 == r for a in A for b in A)
      for r in range(29)]
assert len(set(A)) == 10
assert set(range(6)) <= set(A)
assert min(c) == 1 and max(c) == 6 and sum(c) == 100
print(c)
```

This is a verification of one specified set. The universal upper bound in Theorem 1 is the preceding injection argument, and does not require enumeration of admissible subsets.

3 Relation to Ruzsa bases and scope

For a modulus m , the Ruzsa number R_m is the least representation cap for which an additive basis of $\mathbb{Z}/m\mathbb{Z}$ exists. Ding et al. define L_m as the greatest distinct progression length among bases satisfying $1 \leq \sigma_A(r) \leq R_m$ for every residue, and prove $L_m \leq R_m$ [1, Section 5]. Their extremal family does not impose minimum cardinality.

The value $R_{29} = 6$ and the basis

$$B = \{0, 1, 2, 3, 4, 6, 10, 17, 22\}$$

are already recorded by Sándor and Yang [2, Appendix] and appear again in Ding et al. [1, Section 2.2]. The displayed basis B has longest progression length five, as checked by examining all $29 \cdot 28$ starts and nonzero steps. The ten-element set (1) attains the endpoint-pair bound with length six. Using the reported equality $R_{29} = 6$, the family defining L_{29} is precisely $\mathcal{F}_6$, and Theorem 1 gives $L_{29} = 6$. We do not independently reprove the optimality computation for R_{29} ; it is unnecessary for the fixed-cap theorem.

The contribution is the explicit attaining set at this modulus. The endpoint-pair bound is prior work. The value five refuted here is not attributed as a conjecture to the cited authors, and this note does not settle their question about equality $L_m = R_m$ for infinitely many moduli. For broad adjacent context on constructions involving sums in finite abelian groups, see Zeng’s work on rainbow trees [3]; no result from that work is used in the proof.

The literature comparison is limited to the inspected primary passages and the bounded arXiv searches recorded on 8 September 2026. Those passages do not display A_* or state $L_{29} = 6$; this observation does not establish priority or exclude an occurrence elsewhere. No classification of all admissible bases is claimed.

Computational and AI assistance. OpenAI Codex assisted with the mathematical analysis, exact verification scripts, source comparison, and manuscript preparation. Exact integer and finite-set calculations checked all 100 ordered pairs of A_* , and a separate verification used reflection intersections $\sigma_A(r) = |A \cap (r - A)|$. An additional AI review pass reconstructed the proof and checked the certificates; it was not external peer review. All $29 \cdot 28$ progression parameters were examined for each of the two displayed sets. No floating-point evidence or exhaustive search over all subsets is needed for the theorem.

References

- [1] Yuchen Ding, Huixi Li, Junfeng Li, Wei Niu and Xiamiao Zhao, *Adjacent comparison bounds and extremal sets for Ruzsa numbers*, arXiv:2606.11069v1 (2026). <https://arxiv.org/abs/2606.11069v1>.
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